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Screening for hereditary fructose intolerance mutations by reverse dot-blot

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An assay is described which is useful for genetic screening of the two most prevalent mutations that cause hereditary fructose intolerance (HFI). Both mutations lie within exon 5 of the aldolase B gene. Amplification of exon 5 from genomic DNA isolated from peripheral lymphocytes using biotinylated aldolase B-specific primers yields a biotin-tagged probe. This probe is hybridized to complementary poly(dT)-tailed allele specific oligonucleotides (ASOs) that are bound to a nylon membrane. The length of the ASOs, the amount bound to the membrane and the time of hybridization are optimized for discrimination of all four alleles under the same hybridization conditions. Detection of biotinylated amplified DNA is performed by creating an avidin-alkaline phosphatase complex and visualization by chemiluminescence. This assay can rapidly detect the two mutations, A149P and A174D, which cause >70% of HFI worldwide, and offers a rapid and sensitive assay that is much less invasive for the diagnosis of this often difficult to diagnose disorder.
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KEYWORDS: reverse dot-blot, polymerase chain reaction, allele specific oligonucleotide, non-radioactive, chemiluminescence, fructose intolerance, aldolase B.

INTRODUCTION

Hereditary fructose intolerance (HFI) is a genetic disorder caused by mutations in the aldolase B gene that encodes the isozyme of fructose 1,6-bisphosphate aldolase found in the liver, kidney and small intestine.[1–3] This disorder manifests itself as an intolerance to fructose containing foods. Those most at risk are infants and newborns, for whom diagnosis is often difficult. In these cases, HFI may lead to serious hypoglycemia, growth retardation and/or death.[4] Adults usually develop an aversion to sweets and avoid fructose containing foods, although they are prone to a lifetime of inadvertent fructose ingestion and the accompanying symptoms of hypoglycemia, nausea and vomiting. Once diagnosed, HFI may be effectively treated through diet modification, and if treated properly, individuals can live a normal life without any chronic effects. If not, however, HFI subjects suffer episodes of hypoglycemia, general ill health and risk of death for the remainder of life.[5]

The two standard methods for diagnosis of HFI are the measurement of clinical symptoms upon intravenous fructose challenge and/or the direct assay of aldolase activity in liver biopsy samples.[6,7] During the newborn period, when repercussions of fructose ingestion are the most severe, the diagnosis of HFI is critical. These invasive methods, however, are often precluded by the acute symptoms suffered during infancy. A rapid, effective and relatively non-invasive test would be ideal for detection and diagnosis of HFI infants.

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Table 1. Frequencies of widespread HFI alleles

Mutation	Number of independent alleles (%)[a]	Reference
A149P	226 (56)	(18)
A174D	62 (16)	(19)
N334K	14 (4)	(20)
Δ4E4	10 (3)	(21)
R59op	3 (1)	(22)
L288ΔC	3 (1)	(19)
Y203oc	3 (1)	(23)
A337V	2 (0·5)	(13)
R303W	2 (0·5)	(17)
Q20ΔA	2 (0·5)	(17)
L256P	2 (0·5)	(24)
Other (private)[b]	12 (3)	(13)
Unidentified	59 (15)	(13)

[a] Compiled from published data[13,14,17] and data from clinical investigations in authors' laboratory.

[b] Includes 12 lesions; C239op, C134R, Δ4IVS2-E3, Δ7 + 1, ΔE4-E5, ΔE6-E7, IVS6sas, IVS5n + 1, M1T, R3op, W147R and V104 + 12.

The human aldolase B gene has been cloned and characterized.[8] It consists of nine exons and those regions that encode the active site are found in exons 3, 5, 6 and 7.[9–12] Twenty-three different mutations that cause HFI have been identified[13] and in some cases mutation screening has been used as an alternative for more invasive diagnostic tests.[14–16] The frequencies of all known alleles have been calculated from published reports[13,14,17] and clinical investigations (D. Tolan, unpubl. data) and are shown in Table 1. Of the 23 HFI mutations, A149P and A174D,[18,19] comprise over two-thirds of the mutant alleles world-wide. In some countries these two mutations comprise well over 95% and screening for them can be diagnostic.[15] In addition, there have been a number of mutations that were once thought to be private and now are shared among unrelated families and associated with particular ethnic groups.[17] This situation is particularly suitable for application of a reverse dot-blot procedure where a number of mutations can be screened simultaneously.

Genetic screening for HFI currently utilizes radioactive allele specific oligonucleotides (ASOs) to detect mutations in PCR-amplified genomic DNA bound to a dot-blot.[16] Although this method is very effective, it is too hazardous and costly for widespread screening or use in most clinical settings. A modified screening method is reported here that uses a reverse dot-blot procedure and probes labelled with biotin for detection by chemiluminescence.

MATERIALS AND METHODS

Subjects

Sixteen HFI chromosomes derived from eight unrelated HFI subjects among a cohort of 65 North American HFI subjects, referred from various physicians and pediatric units, were examined. The genotypes of all subjects had been determined previously by standard dot-blot and ASO hybridization procedures.[16] Blood samples from these individuals were collected in EDTA and the DNA was extracted according to standard protocols.[16]

Oligonucleotide synthesis and preparation of the reverse dot-blot membranes

Biotinylated oligonucleotides were prepared on a Cyclone or Milligen Model 7500 DNA Synthesizer using phosphoramidite chemistry and the manufacturer's protocols. Poly(dT)-tailing of ASOs used the method of Kawasaki *et al.*[25] Briefly, ASOs (10–200 pmol) were heated to 70°C for 15 min then cooled to 4°C for 1 min prior to incubation at 37°C in a buffer containing 100 mM potassium cacodylate, 25 mM Tris-HCl, pH 7·6, 1 mM $CoCl_2$, 0·2 mM dithiothreitol, 2 mM dTTP and 60 units of terminal deoxyribonucleotidyl transferase (TdT) (Gibco BRL, Gaithersburg, MD, USA). The tailing reactions were monitored in some cases using 1:1000 parts α-[^{35}S]-dATP (30 μCi). The reactions were stopped by addition of 5 mM EDTA and heating to 65°C for 15 min. The tailed ASOs were diluted with 10 mM Tris-HCl, 1 mM EDTA, pH 7·6 and applied to a nylon membrane (Biodyne B Membranes, Gibco BRL, Gaithersburg, MD, USA) by vacuum filtration using a dot-blot apparatus and the manufacturer's protocol (Bio-Rad, Hercules, CA, USA). After filtration the membrane was washed with 6X SSC (1X is 0·15 M NaCl, 15 mM sodium citrate, pH 7) allowed to air dry, treated with ultraviolet light at 254 nm for 5·5 s to cross-link the tailed ASOs to the membrane, and dried in a vacuum oven for 30 min at 80°C.

PCR amplification

Mutant cloned DNA[16] (5–10 ng) or patient DNA (0·2–1·5 μg) was used for each reaction (50 μl), which contained 50 mM KCl, 10 mM Tris-HCl, pH 8·4, 1·5–2.0 mM $MgCl_2$, 0·2 mM oligonucleotide primers, 0·2 mM each dNTP (dATP, dGTP, dCTP and dTTP), and 1·25 units of *Taq* DNA Polymerase (Perkin-Elmer, Foster City, CA, USA). DNAs were heated to 94°C

for 5 min prior to addition of the cocktail. The amplification was done in a thermocycler (Perkin-Elmer, Foster City, CA, USA) as follows: 5 cycles of 94°C for 1·5 min, 55°C for 1 min, 72°C for 1 min; 25 cycles of 94°C for 1·5 min, 55°C for 1 min, 72°C for 3 min; and finally 7 min at 72°C then a 30 min ramp to 4°C.

Hybridization and washing of the membranes

Membranes with bound ASOs were sealed in a plastic bag containing a solution (0·3–0·5 ml cm^{-2}) of 4X SSPE/0·5% SDS at 42°C for 30 min (1X SSPE is 0·15 M NaCl, 10 mM NaH_2PO_4, 1 mM EDTA, pH 7·4). Polymerase chain reaction-amplified probe DNA (200 ng) was denatured by mixing with 0·2 M NaOH, 5 mM EDTA in 100 μl, heating the mixture at 95°C for 10 min, and quickly cooling to 4°C for 2 min. Freshly denatured probe DNA was added directly to the prehybridization solution in the plastic bag, sealed and incubated at 37–57°C. The membranes were washed three times in 1X SSPE/0·2% SDS at 25°C for 5 min.

Detection of the signals

The detection of hybridized biotinylated probes was performed by means of a chemiluminescent reaction at 25°C. The membranes were washed in 1X blocking buffer (5% SDS, 17 mM Na_2HPO_4, 8 mM NaH_2PO_4) for 5 min, then incubated in 1X blocking buffer plus streptavidin (1 μg ml^{-1}) (Sigma, St. Louis, MO, USA) for 5 min. The membranes were washed three times in 0·1X blocking buffer, then incubated in 0·1X blocking buffer plus biotinamidocaproyl-labelled alkaline phosphatase (1 μg ml^{-1}) (Sigma, St. Louis, MO, USA) for 5 min. Chemiluminescence was developed by rinsing three times in wash buffer (10 mM Tris, 10 mM NaCl, 1 mM $MgCl_2$) then incubated in 0·02 ml cm^{-2} of a 0·25 mM 1,2-dioxetane chemiluminescent substrate for alkaline phosphatase, CSPD® (Tropix, Bedford, MA, USA), for 10 min at 37°C. After development, membranes were autoradiographed for 15–30 min.

RESULTS

PCR amplification

The oligonucleotide primers (Table 2) used in the PCR amplification flank exon 5 and produce a fragment of 336 bp. The PCR-amplified fragments generated using a biotin labelled 3′-primer (PCR-2B) are complementary to the ASOs that were fixed to the nylon membrane. The PCR amplification products were routinely examined on an agarose gel as an assay for the amplification reaction (Fig. 1).

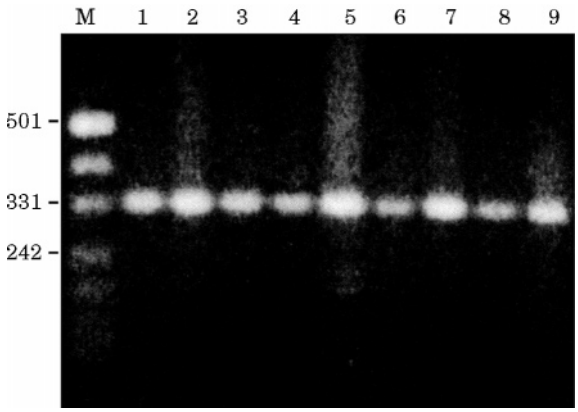


Fig. 1. Electrophoresis on 1% agarose gel of polymerase chain reaction (PCR) amplified genomic DNAs with exon 5 primers, E5 and PCR-2B (a biotinylated primer). M, molecular weight markers, lane 1 was generated from a clone with the wild-type sequence, lanes 2–9 were generated from patient DNA samples. Each lane was from 10% of the PCR reaction. Electrophoresis was for 45 min at 100 V prior to staining in ethidium bromide.

Reverse dot-blot

A pair of ASOs (mutant and wild-type) that differ at a single-bp discriminate between their cognate target DNAs by hybridization under stringent conditions. So that a single hybridization step could be used, a pair of ASOs should discriminate under the same conditions; these conditions should be discriminatory for other pairs of ASOs that are specific for other mutations such that a panel of mutations can be tested simultaneously. The length of the ASOs for one set of alleles was modified to match the discrimination temperature of another set. Several pairs of ASOs for A174D were synthesized in attempts to match the discrimination temperature for the A149P ASOs (Table 2). These A174D ASOs were tested for their discriminatory temperature using radiolabelled ASOs in the normal dot-blot procedure against known wild-type and mutant DNA sequences bound to membranes.[16] The pair A174wt.64/A174D.64 worked at the same temperature as the pair used for A149P.

For each of the two mutations, A149P and A174D, the four ASOs (A149wt, A149P, A174wt.64 and A174D.64) were extended with a poly(dT)-tail. Poly(dT)-tails of different lengths were generated by incubation with the TdT for 1, 3 and >12 h. The various lengths of poly(dT)-tailed ASO were tested for optimal binding and hybridization conditions by

Table 2. Sequence of the oligonucleotides used for amplification and dot-blotting

Use/ oligonucleotide	5′→3′	Calculated T_d °C[a]	Discriminatory temperature °C[b]
PCR amplification:			
E5	ACTCCTTCCCTTTATTA	—	—
PCR-2B	B[c]-GGTCCATTTGTAGTTATAGT	—	—
ASOs:			
A149wt	AAGTGGCGT**G**[d]CTGTGCTGA	62	67
A149P	AAGTGGCGT**C**CTGTGCTGA	62	67
A174wt	TCGCTACG**C**CAGCATCT	56	58
A174D	AGATGCTG**T**CGTAGCGA	54	56
A174wt.64	TCGCTACG**C**CAGCATCTGT	62	67
A174D.62	AGATGCTG**T**CGTAGCGAGC	62	64
A174D.62.2	CTCGCTACG**A**CAGCATCTG	62	64
A174D.64	TCGCTACG**A**CAGCATCTGT	60	67

[a] Calculated by rule; $T_d = 2°C$ (A/T) + 4°C (g/C) + 2°C (26).
[b] Discrimination determined by normal dot-blot procedures and is only an assurance of matching conditions, not the reverse dot-blot hybridization temperature.
[c] B; biotin.
[d] **Bold type** indicates the site of a mismatch due to a mutation.

screening with complementary radiolabelled oligonucleotides. ASOs tailed in overnight reactions showed the strongest and most stable binding to the nylon membrane (data not shown).

Radiolabelled poly(dT)-tailed ASOs bound to nylon membranes were exposed to different amounts of u.v. light to determine the optimal amount for cross-linking. The label was quantified using a STORM860 phosphorimager (Molecular Dynamics, Sunnyvale, CA, USA), which showed optimal binding occurred at 100 mWatts cm^{-2} (data not shown). Different amounts (0·1–100 ng) of the tailed ASOs were tested for hybridization to PCR generated complementary biotinylated probes and their ability to discriminate between wild-type and mutant DNA sequences. The optimal amount of mutant tailed ASO that gave a strong signal with mutant PCR-amplified probe and little signal with the wild-type PCR-amplified probe and vice versa was 10–50 ng/dot (data not shown).

Hybridization of biotinylated PCR-amplified probe to poly(dT)-tailed ASOs was tested by varying the hybridization solution, time and temperature. A 4X SSPE/0·5% SDS solution eliminated most background noise without affecting the hybridization signal, whereas standard solutions containing Denhardt's[27] gave a poorer signal and a higher background. The length of time for hybridization was tested at 0·5, 1·0, 3·0 and <12 h. When incubated for only 0·5 or 1·0 h, probe signals were low and faint. Optimal binding occurred within 3·0–18 h of hybridization and was used in all subsequent hybridization experiments. The hybridization temperature ranged from 37 to 57°C, with discriminate binding of PCR-amplified biotinylated DNA to the poly(dT)-tailed ASOs at 57°C.

A reverse dot-blot containing the four ASOs for screening the A149P and A174D mutations was probed with the PCR-amplified biotin labelled DNA from individuals that harbour these alleles using the optimal conditions described above (Fig. 2). Three compound heterozygote individuals were screened. One individual was A149P/A174D, another was A174D/wild-type at A149, and the third was A149P/wild-type at A174. The different genotypes determined by reverse dot-blot were in accordance with the previously determined genotypes by means of classical dot-blot assays.[16]

DISCUSSION AND CONCLUSIONS

The cloning and DNA sequence determination of the aldolase B gene[8] has made possible the use of genetic screening methods for the diagnosis of this difficult to diagnose disorder.[6] Often HFI subjects present with acute symptoms that are indistinct from a variety of other disorders and their acute illness often precludes the invasive methods that would normally be required for proper diagnosis. A genetic screen, based on a sample of blood, would be an excellent alternative, and indeed would be ideal in a newborn screening programme if the cost could be sufficiently reduced. The development of such a screening test would require: (1) that a sufficient percentage of mutations that are known to cause HFI could be screened such that the test could be diagnostic (>90%); and (2) that a screening test could be easily performed in any clinical laboratory setting. The development of the reverse dot-blot described here begins to solve these

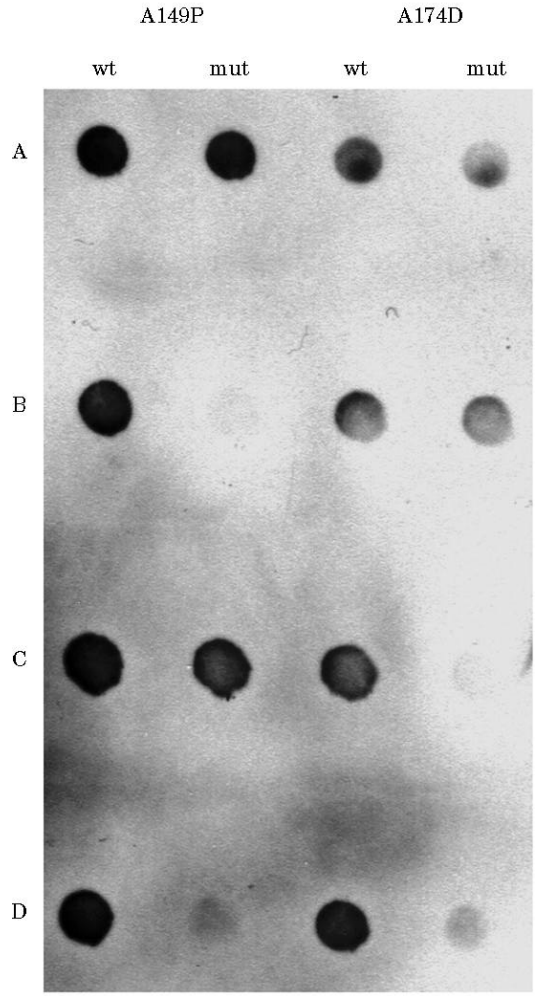


Fig. 2. Reverse dot-blot analysis of individuals with known genotypes. Hybridization of polymerase chain reaction (PCR) fragments amplified from exon 5 as shown in Fig. 1 using primers E5 and PCR-2B. (A) A149P/A174D, (B) A174D/wild-type, (C) A149P/wild-type, (D) wild-type control. Hybridization was for 18 h at 57°C to blots with tailed ASOs (17 pmol).

problems. This method can simultaneously test for a number of different mutations. Indeed it has been used to test for 11 CF mutations.[28] Currently, 11 of all known HFI mutations are widespread and account for >82% of HFI alleles (Table 1), although this percentage largely applies to the European population. If and when more mutations in HFI subjects worldwide are identified, the reverse dot-blot such as described here could easily be modified to include more mutations. Also, the reverse dot-blot assay can be performed in virtually any clinical laboratory setting due to the elimination of radioactive reagents necessary when using other methods.[16] Other non-radioactive methods have been reported,[14] but these methods are not as easily modified to include more alleles. All that is required are the biotinylated PCR primers and reverse dot-blot membranes containing the ASOs for the mutations to be screened.

Twenty-three lesions have been identified in the aldolase B gene that cause HFI. Most of these mutations are either private or comprise less than 1% of the HFI chromosomes (Table 1); however, the reverse dot-blot described here includes the two most common mutations, which comprise 72% of HFI chromosomes. There are five other mutations that comprise 1–4% of alleles (Table 1). Inclusion of these five in a reverse dot-blot assay would increase the percentage of detectable mutations from 72 to 80%, but would require multiplex amplifications of exons 3, 4, 5, 6, 8 and 9 and the optimization of 10 other ASOs.[19–23] Eventually, this will be necessary if over 90% of HFI mutations were included in an assay.[13] However, this is not currently possible on a world-wide basis. A significant percentage of HFI alleles, perhaps common in non-European populations, remain unidentified. Until these issues are resolved, the reverse dot-blot described here is sufficient for detection of 72% of HFI mutations world-wide, and sufficient for >90% of HFI alleles in the United Kingdom[15] and France[14] and is an extremely helpful tool in the important and critical diagnosis of children at risk. This genetic test is quick and sensitive and can be performed within 2 days, or even less if the DNA is more readily obtained, for example from dried blood spots.[29,30]

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